



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations

Module #4

**Effective Multiplication Factor K_{eff} ,
the Neutron Life Cycle,
Criticality and Reactivity**

**Dept. of Nuclear Engineering
North Carolina State University**

NE235 Nuclear Reactor Operations Training **NRP**

Module #4

Module #4 Contents:

1. “Sources & Sinks” of neutrons in reactor
2. 6 Factor Formula: K-effective (K_{eff})
3. Neutron Lifecycle
4. Reactivity (ρ) & Delayed Neutron Fraction β

Laboratory Session this week:

- Lab #3 - PULSTAR Reactor Startup

Homework #4 – Due 9/27

NE235 Nuclear Reactor Operations Training

Module #4

Review

1. Nuclear Reactions in reactor core involving neutrons

- Absorption reactions
- Scattering reactions
- Fission Process

2. Elemental Components of Core Structures

3. Reaction Rate Calculations

So, now we understand the elemental compositions of the core structures, the properties of the isotopes that make up those elements, the kinds of nuclear reactions those isotopes can undergo with different energy neutrons, and what kinds of products result from those reactions (radioactive isotopes, radiation, and energy). This forms a basis for calculating reaction rates (e.g. fission, etc...) and understanding what is happening to neutron population in the core.

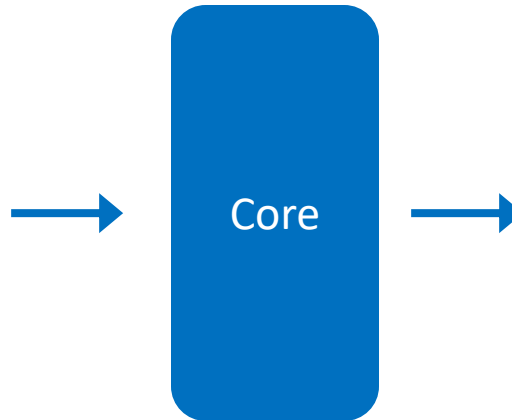
Sources & Sinks of Neutrons in Reactor

"Sources" (add neutrons)

- * prompt & delayed fission neutrons from fast & thermal fission

η "eta"
 ϵ "epsilon"

+ "source reactions"



"Sinks" (remove neutrons)

- * absorption inside core
- * leakage out of core (as fast or thermal)

$\mathcal{L}_F$
 p
 $\mathcal{L}_{TH}$
 f

Sources & Sinks of Neutrons in Reactor

□ Sources – Add neutrons to fission cycle (η , ϵ):

i. Fission N_f (e.g. of ^{235}U , ^{239}Pu , ^{238}U)

Thermal (η) and Fast (ϵ) Fission

- Prompt Neutrons - 99.3% of ^{235}U fission neutrons are prompt (*prompt generation time* $\ell_p \sim 3 \times 10^{-5} \text{ sec}$)
- Delayed Neutrons – 0.7% of ^{235}U fission neutrons are delayed (*delayed generations time* $\ell_d \sim 12.7 \text{ sec}$)

ii. “Source Neutrons” S :

- Spontaneous Fission (e.g. of ^{238}U , ^{235}U)
- “Photo-neutrons”: $^2\text{H}(\gamma, n)$ & $^9\text{Be}(\gamma, n)$
- $^{18}\text{O}(\alpha, n)$ reaction in fuel (UO_2)
- PuBe Source $^{239}\text{Pu}(\alpha) \rightarrow ^9\text{Be}(\alpha, n)$

□ Sinks – Remove neutrons from fission cycle ($\mathcal{L}_F$, p , $\mathcal{L}_{TH}$, f):

i. Leak out of Reactor:

- As Fast ($\mathcal{L}_F$)
- As Thermal ($\mathcal{L}_{TH}$)

ii. Absorbed in Non-Fuel:

- In Control Rods (f)
- In Moderator (f)
- In $^{238}\text{U} \Leftrightarrow$ Doppler Feedback Effect (p)
- In Fission Product Poisons (f)
- In other core structures (cladding, experiments, etc...) (f)

Where:

$$S \ll N_f \text{ when } k_{\text{eff}} \rightarrow 1$$

6 Factor Formula & K-effective

$$K_{\text{eff}} = \eta \epsilon \mathcal{L}_F p \mathcal{L}_{\text{TH}} f$$

>1 | Supercritical

K_{eff} $=1$ | Critical

<1 | Sub-Critical

η “eta” = Reproduction Factor =

$$\frac{\# \text{ fission neutrons from Thermal Fission}}{\# \text{ thermal } n\text{'s absorbed in Fuel}}$$

- $\eta \sim 2.43$ for ^{235}U prompt + delayed
- $\sim 5/6$ neutrons absorbed cause fission
- $\sim 1/6$ neutrons captured in ^{235}U without fissioning

ϵ “epsilon” = Fast Fission Factor =

$$\frac{\# n\text{'s from Fast \& Thermal Fission}}{\# n\text{'s from Thermal Fission}}$$

- Fissionable Isotopes $\Rightarrow ^{238}\text{U}$
- $\epsilon > 1$

6 Factor Formula & K-effective

$$K_{\text{eff}} = \eta \epsilon \mathcal{L}_F p \mathcal{L}_{\text{TH}} f$$

$\mathcal{L}_F$ = **Fast Non-Leakage Probability** = $\frac{\text{\# Fast Neutrons that start to slow down \& moderate}}{\text{Total \# n's from Fast \& Thermal Fission}}$

- Number that don't leak = $\mathcal{L}_F$ (<1)
- Number that leak = $1 - \mathcal{L}_F$

p = **Resonance Escape Probability** = $\frac{\text{\#n's reaching Thermal Energy}}{\text{\#n's that started slowing down and moderating}}$

- Number that are captured in resonances of ^{238}U = $1 - p$; ($p < 1$)

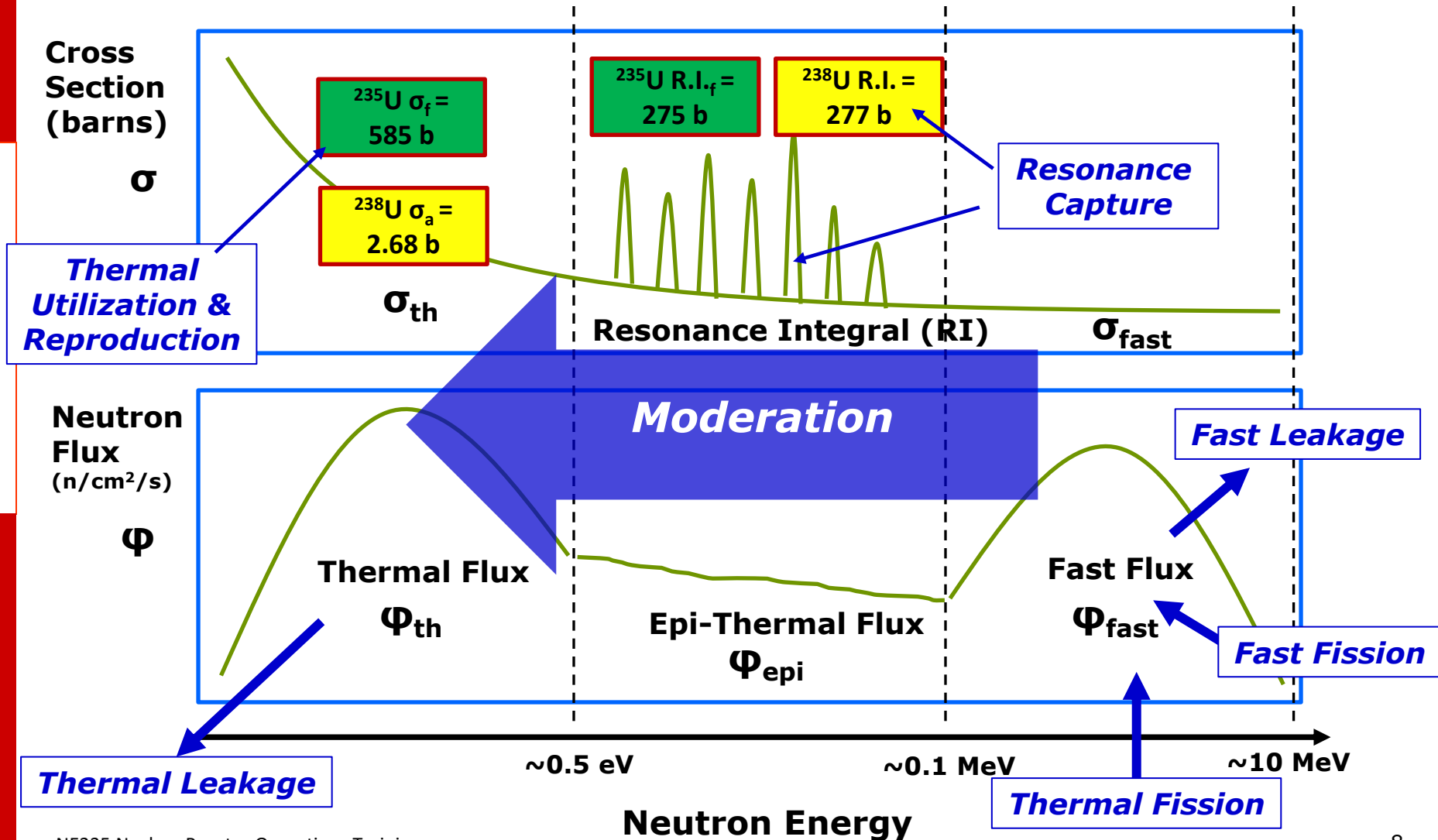
$\mathcal{L}_{\text{TH}}$ = **Thermal Non-Leakage Probability** = $\frac{\text{\# Thermal n's Absorbed in Core}}{\text{\#n's that reach Thermal Energy}}$

- Number that leak = $1 - \mathcal{L}_{\text{TH}}$

f = **Thermal Utilization Factor** = $\frac{\text{\# Thermal n's Absorbed in Fuel } (^{235}\text{U})}{\text{\# Thermal n's Absorbed in Core}}$

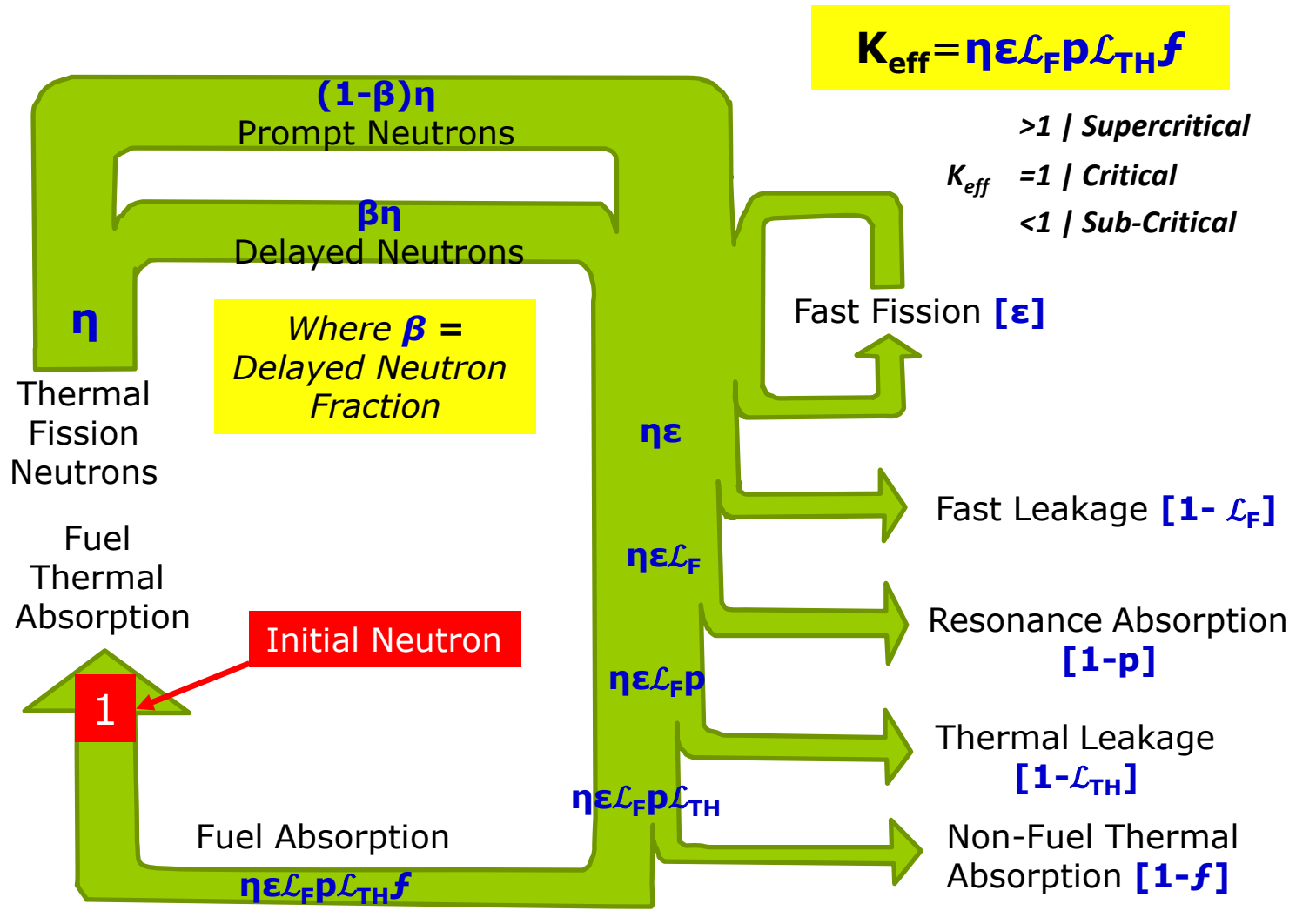
- Number that are absorbed in fuel (e.g. ^{235}U) –vs– # absorbed in fuel & core materials

Neutron Energy Spectrum and energy dependent reaction Cross Section



Neutron Lifecycle

10 MeV
5 MeV
2 MeV
1 MeV
0.5 MeV
0.2 MeV
0.1 MeV
10 eV
1 eV
0.1 eV
0.01 eV



Reactivity & Delayed Neutron Fraction

Reactivity: Departure from “Criticality” ($K_{\text{eff}}=1$) due to mismatch between sources and sinks:

$$\rho = (K_{\text{eff}} - 1)/K_{\text{eff}}$$

> 0 Super Critical	$(K_{\text{eff}} > 1)$
$= 0$ Critical (Delayed Critical)	$(K_{\text{eff}} = 1)$
< 0 Sub-Critical	$(K_{\text{eff}} < 1)$

Units: $\rho(\Delta K/K)$ where $0.00001 \Delta K/K = 0.001\% \Delta K/K = 1\text{pcm}$ (percent milli-rho)

Delayed Neutron Fraction β :

A reasonable value for the **Delayed Neutron Fraction β** of total neutrons in the PULSTAR reactor is $\beta = 0.007$ (for ^{235}U fission). **$\sim 0.7\%$ of fission neutrons arise from the decay of neutron rich fission product isotopes** which decay with short half lives.

Effective Delayed Neutron Fraction β_{eff} :

Delayed neutrons are “born” at lower energies than prompt neutrons and are therefore less likely to leak out of the reactor as fast neutrons or to be absorbed in the resonance region. **This makes them more “effective” in propagating the chain reaction.** Thus, the **Effective Delayed Neutron Fraction, $\beta_{\text{eff}} = 0.0073$** (for ^{235}U fission).

Delayed Neutron Fraction, β

Boundary Between Delayed and Prompt Critical:

A reactor is said to be delayed critical when production of prompt fission neutrons plus delayed neutron precursors is just equal to the loss rate of neutrons by absorption and leakage.

ρ (pcm)	Status	ρ/β (Dollars)
> 730	Super Prompt Critical	> \$1
=730	Prompt Critical	= \$1
$0 < \rho < 730$	Delayed Super Critical	$0 < \rho/\beta < 1$
= 0	Delayed Critical	= 0
< 0	Sub Critical	< 0

If the production rate of prompt neutrons is just equal to the loss rate of neutrons, the reactor is critical on prompt neutrons alone ($\rho = 730$ pcm; $\rho/\beta = \$1$); it doesn't have to wait for the delayed neutrons to maintain the chain reaction!

Summary

- **Sources & Sinks** – *thermal and fast fission –vs- absorption and leakage.*
- **6 Factor Formula:** $K_{eff} = 1 \rightarrow$ *Critical = constant neutron population sustained by fission!*
- **Neutron Lifecycle:** *prompt ($1-\beta$) and delayed (β) fission occur over different timeframes.*
- **Reactivity (ρ) & ρ/β :** *Departure from criticality relative to delayed neutron fraction determines how fast neutron pop. changes!*